

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

19th June 2019

**Subject CM2B – Financial Engineering and Loss
Reserving (Paper B)**

Time allowed: 1 Hour 45 Minutes (14.45 – 16.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Mark allocations are shown in brackets.***
- 2. Attempt all questions, beginning your answer to each question in the template provided.***
- 3. Attempt all sub-parts of the question in the template provided only, unless otherwise instructed to do so.***
- 4. The working of each part of the question should be on a separate tab (sheet). For example, question 1(i) should be worked out within the tab (sheet) with name 1(i) of worksheet named CM2B.***
- 5. Where possible, summarize your data used & assumptions made (if any) in a separate tab.***
- 6. Do save your work in Excel files on a regular basis.***
- 7. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.***
- 8. Ensure that you click the “Submit” button only when you have completed the question paper and final submission has to be made.***

AT THE END OF THE EXAMINATION

Please return this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you. You are requested to save and submit the work before leaving the examination premises.

- Q. 1)** A well-known airline company has run into financial difficulty recently. The current share price of the Company is Rs 200/- per share. Two investors (Investor A and Investor B) want to use the opportunity to make extra-ordinary profits. They plan to buy few shares now and sell those shares one year from now. Assume that the risk-free interest rate is 'Zero'.

The investors after conducting some research conclude the following share prices one year from now.

Probability of event happening (1)	Share Price- one year from now
20%	0
10%	50
30%	150
25%	320
5%	400
10%	500

- i) Find the following measures for a period of next one year:
- a) Mean of the return (2)
 - b) Variance of the return (2)
 - c) Downside Semi-variance of the return (3)
 - d) Value at Risk at 10% significance (3)

Both the investors have an initial wealth of 2 lakh rupees each. Each plan to buy 1000 shares of the company.

Investor 'A' has a utility function of $U(w) = w^2$, where w is the level of wealth.

- ii) Plot a chart of $U(w)$ against values of w as per column 2 of the above table. (4)
- iii) Find the Mean of the Utility function and the Utility function of the Mean wealth. (4)

Investor 'B' has a utility function of $U(w) = w^{1/4}$, where w is the level of wealth.

- iv) Plot a chart of $U(w)$ against values of w as per column 2 of the above table. (4)
- v) Find the Mean of the Utility function and the Utility function of the Mean wealth. (4)
- vi) Find the Certainty equivalent of wealth for Investor 'A' and Investor 'B'. Comment on the attitude towards risk of Investor 'A' and Investor 'B' separately based on Certainty equivalents. Who would buy the Shares? (8)

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- Q. 2)** A general insurance company is selling three type of additional riders ADB, TPD and TI. The company has received a total of 5326 claims in these three riders during the year 2005 – 2013. The details of each claim reported to the company is provided in the data sheet.

- i) Determine the average claim size reported to the company under the three riders. (6)
- ii) Determine the amount of TPD claims for each accident year in each development year. (10)

- iii) Determine the development factor at each duration using the Chain Ladder method. (5)
- iv) Complete the TPD Triangles using the Chain Ladder method. (5)
- v) Calculate the Ultimate and IBNR claims against each accident year. (5)

Candidates are requested to clearly mention the assumptions, used in each step.

Data in spreadsheet provided.

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- Q. 3)** A share is currently priced at Rs. 640. A writer of 1000 units of a one year European put option with an exercise price of Rs. 630 has delta-hedged the option with a portfolio which holds cash and is short 250 shares. The continuously compounded risk-free rate of interest is 3% p.a. and no dividends are payable during the life of the option.

The assumptions of the Black-Scholes model apply.

- i) Calculate the delta of the option. (4)
- ii) Calculate the volatility of the share implied by the delta. (9)
- iii) a) Calculate the price of the option. (4)
b) Determine the value of the cash holding in the hedging portfolio. (2)
- iv) Using put-call parity, calculate the price of the corresponding call-option using the price of the put option calculated in part (iii). (3)
- v) Calculate the price of call option and the call pay-off for the range of share price given in the worksheet with other parameters same as mentioned above. (10)
- vi) Prepare the chart of the call option price and the call pay-off against the Share price. (3)

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Link for Data Set: http://www.actuariesindia.org/downloads/CM2B_Data.xlsx
